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R Programming
ITA0401

1. Write an R program to create a vector containing the marks of 10 students. Display the vector and print the total number of elements in the vector.

Output :

```
marks <- c(85, 90, 78, 88, 92, 76, 81, 95, 89, 84)
> print("Marks of 10 Students:")
[1] "Marks of 10 Students:"
> print(marks)
[1] 85 90 78 88 92 76 81 95 89 84
>
> print("Total number of elements in the vector:")
[1] "Total number of elements in the vector:"
> print(length(marks))
[1] 10
```

2. Write an R program to create two vectors containing the monthly sales of two different stores.

Output :

```
store1 <- c(12000, 15000, 18000, 20000, 17000, 16000)
> store2 <- c(10000, 14000, 16000, 19000, 15000, 14000)
> add <- store1 + store2
>
> sub <- store1 - store2
>
> mul <- store1 * store2
```

```
>
> div <- store1 / store2
>
> print("Store 1 Sales:")
[1] "Store 1 Sales:"
> print(store1)
[1] 12000 15000 18000 20000 17000 16000
>
> print("Store 2 Sales:")
[1] "Store 2 Sales:"
> print(store2)
[1] 10000 14000 16000 19000 15000 14000
>
> print("Addition:")
[1] "Addition:"
> print(add)
[1] 22000 29000 34000 39000 32000 30000
>
> print("Subtraction:")
[1] "Subtraction:"
> print(sub)
[1] 2000 1000 2000 1000 2000 2000
>
> print("Multiplication:")
[1] "Multiplication:"
> print(mul)
[1] 1.20e+08 2.10e+08 2.88e+08 3.80e+08 2.55e+08 2.24e+08
>
```

```
> print("Division:")
[1] "Division:"
> print(div)
[1] 1.200000 1.071429 1.125000 1.052632 1.133333 1.142857
```

3. Write an R program to create a vector of 8 employee salaries. Find and display:

Output :

```
# 8 employee salaries
>
> salary <- c(25000, 30000, 28000, 35000, 40000, 32000, 27000, 36000)
>
> max_salary <- max(salary)
>
> min_salary <- min(salary)
>
> total_salary <- sum(salary)
>
> average_salary <- mean(salary)
>
> print("Employee Salaries:")
[1] "Employee Salaries:"
> print(salary)
[1] 25000 30000 28000 35000 40000 32000 27000 36000
>
> print("Maximum Salary:")
[1] "Maximum Salary:"
> print(max_salary)
[1] 40000
>
> print("Minimum Salary:")
[1] "Minimum Salary:"
> print(min_salary)
[1] 25000
>
> print("Total Salary:")
[1] "Total Salary:"
> print(total_salary)
[1] 253000
>
> print("Average Salary:")
[1] "Average Salary:"
```

```
> print(average_salary)
[1] 31625
```

```
>
```

4. Write an R program to create a vector containing the ages of 12 employees. Display the vector

Output :

```
> # ages of 12 employees
>
> ages <- c(25, 32, 28, 45, 30, 27, 40, 35, 29, 38, 31, 26)
> ascending <- sort(ages)
> descending <- sort(ages, decreasing = TRUE)
>
> print("Employee Ages:")
[1] "Employee Ages:"
> print(ages)
[1] 25 32 28 45 30 27 40 35 29 38 31 26
>
> print("Ascending Order:")
[1] "Ascending Order:"
> print(ascending)
[1] 25 26 27 28 29 30 31 32 35 38 40 45
>
> print("Descending Order:")
[1] "Descending Order:"
```

```
> print(descending)
[1] 45 40 38 35 32 31 30 29 28 27 26 25
```

```
>
```

5. Write an R program to create a vector of 15 product IDs. Read a product ID from the user and check whether it exists in the vector. Display an appropriate message.

Output :

```
> # vector of 15 product IDs
>
> product_ids <- c(101, 102, 103, 104, 105, 106, 107, 108,
+                 109, 110, 111, 112, 113, 114, 115)
>
> id <- as.numeric(readline("Enter Product ID: "))
Enter Product ID:
>
> if (id %in% product_ids) {
+   print("Product ID found.")
+   print(paste("Position:", which(product_ids == id)))
+ } else {
+   print("Product ID not found.")
+ }
[1] "P"
```

6. Write an R program to create a vector containing 20 integers. Separate and display

Output :

```

> # vector of 15 product IDs
>
> product_ids <- c(101, 102, 103, 104, 105, 106, 107, 108,
+               109, 110, 111, 112, 113, 114, 115)
>
> id <- as.numeric(readline("Enter Product ID: "))
Enter Product ID:
>
> if (id %in% product_ids) {
+   print("Product ID found.")
+   print(paste("Position:", which(product_ids == id)))
+ } else {
+   print("Product ID not found.")
+ }
[1] "Product ID not found."
> # Containing 20 integers
>
> numbers <- c(12, 25, 18, 33, 40, 51, 62, 77, 84, 95,
+             16, 29, 38, 47, 50, 63, 72, 81, 90, 11)
>
> even_numbers <- numbers[numbers %% 2 == 0]
>
> odd_numbers <- numbers[numbers %% 2 != 0]
>
> print("Original Vector:")
[1] "Original Vector:"
> print(numbers)
[1] 12 25 18 33 40 51 62 77 84 95 16 29 38 47 50 63 72 81 90 11

```

```

>
> print("Even Numbers:")
[1] "Even Numbers:"
> print(even_numbers)
[1] 12 18 40 62 84 16 38 50 72 90
>
> print("Odd Numbers:")
[1] "Odd Numbers:"
> print(odd_numbers)
[1] 25 33 51 77 95 29 47 63 81 11

```

7. Write an R program to create a vector containing the prices of 10 products. Increase each product price by 10% and display both the original and updated price vectors.

Output :

```

# Display the updated prices
> print("Updated Prices (10% Increase):")
[1] "Updated Prices (10% Increase):"
> print(updated_prices)
[1] 110 275 165 330 550 495 220 385 440 660

```

```

>

```

8. Write an R program to create a vector of 15 city names. Display

Output :

```

# 15 city names
>
> cities <- c("Delhi", "Mumbai", "Chennai", "Hyderabad", "Bengaluru",
+           "Pune", "Kolkata", "Jaipur", "Lucknow", "Visakhapatnam",
+           "Ahmedabad", "Bhopal", "Nagpur", "Goa", "Mysore")
>
> print("First Five Cities:")

```

```

[1] "First Five Cities:"
> print(cities[1:5])
[1] "Delhi"    "Mumbai"   "Chennai"  "Hyderabad" "Bengaluru"
>
> print("Last Five Cities:")
[1] "Last Five Cities:"
> print(cities[11:15])
[1] "Ahmedabad" "Bhopal"    "Nagpur"    "Goa"       "Mysore"
>
> print("Cities from Index 6 to 10:")
[1] "Cities from Index 6 to 10:"
> print(cities[6:10])
[1] "Pune"      "Kolkata"   "Jaipur"    "Lucknow"    "Visakhapatnam"
>
>
> print("Every Alternate City:")
[1] "Every Alternate City:"
> print(cities[seq(1, length(cities), by = 2)])
[1] "Delhi"    "Chennai"   "Bengaluru" "Kolkata"   "Lucknow"   "Ahmedabad" "Nagpur"    "Mysore"

>

```

9. Write an R program to create a vector containing the marks of 20 students. Calculate and Display

Output :

```

# marks of 20 students
>
> marks <- c(78, 85, 92, 67, 88, 74, 81, 95, 69, 84,
+           76, 90, 72, 86, 80, 91, 65, 79, 87, 93)
>
>
> mean_marks <- mean(marks)
> median_marks <- median(marks)
> std_dev <- sd(marks)
> variance <- var(marks)
> marks_range <- range(marks)
>
> print("Student Marks:")
[1] "Student Marks:"
> print(marks)
[1] 78 85 92 67 88 74 81 95 69 84 76 90 72 86 80 91 65 79 87 93
>
> print("Mean:")
[1] "Mean:"
> print(mean_marks)

```



```

[1] 81.6
>
> print("Median:")
[1] "Median:"
> print(median_marks)
[1] 82.5
>
> print("Standard Deviation:")
[1] "Standard Deviation:"
> print(std_dev)
[1] 8.987711
>
> print("Variance:")
[1] "Variance:"
> print(variance)
[1] 80.77895
>
> print("Range:")
[1] "Range:"
> print(marks_range)
[1] 65 95

```

```
>
```

10. Write an R program to create a vector containing the performance scores of 15 employees.

Perform the following operations:

Output :

```

# scores of 15 employees
>
> scores <- c(85, 72, 91, 48, 66, 95, 38, 80, 88, 54,
+           76, 99, 45, 83, 69)
>
> above_80 <- scores[scores > 80]
>
> below_50 <- sum(scores < 50)
>
>
> highest <- max(scores)
> lowest <- min(scores)
>

```

```

>
> average <- mean(scores)
> descending <- sort(scores, decreasing = TRUE)
>
>
> print("Performance Scores:")
[1] "Performance Scores:"
> print(scores)
[1] 85 72 91 48 66 95 38 80 88 54 76 99 45 83 69
>
> print("Employees Scoring Above 80:")
[1] "Employees Scoring Above 80:"
> print(above_80)
[1] 85 91 95 88 99 83
>
> print("Number of Employees Scoring Below 50:")
[1] "Number of Employees Scoring Below 50:"
> print(below_50)
[1] 3
>
> print("Highest Score:")
[1] "Highest Score:"
> print(highest)
[1] 99
>
> print("Lowest Score:")
[1] "Lowest Score:"
> print(lowest)
[1] 38
>
> print("Average Performance Score:")
[1] "Average Performance Score:"
> print(average)
[1] 72.6
>
> print("Scores in Descending Order:")
[1] "Scores in Descending Order:"
> print(descending)
[1] 99 95 91 88 85 83 80 76 72 69 66 54 48 45 38
>

```

